INTRODUCTION TO ARTIFICIAL INTELLIGENCE CENG455

Classification Evaluation Metrics &

The Effect Of The Approaches Used On The Performance Of The Classifiers İn The Case Of Selecting Training And Test Sets

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1.INTRODUCTION

Classification evaluation metrics are used to measure the performance of a classifier, which is a machine learning model that is used to assign a class label to a given input data point. There are different types of classification tasks, such as binary classification, where the classifier assigns one of two class labels to the input data, and multi-class classification, where the classifier assigns one of several class labels to the input data. In this article, we will discuss several common evaluation metrics that can be used to evaluate the performance of a classifier, including accuracy, precision, recall, and F1 score. We will also explore how these metrics can be applied to both binary and multi-class classification tasks and discuss the tradeoffs and considerations involved in using them.

In addition to accuracy, precision, recall, and F1 score, there are other evaluation metrics that can be used to measure the performance of a classifier. For example, the receiver operating characteristic (ROC) curve is a graphical plot that illustrates the performance of a binary classifier as the classification threshold is varied. The ROC curve is created by plotting the true positive rate (TPR) on the y-axis and the false positive rate (FPR) on the x-axis. The TPR is calculated as the number of true positive predictions divided by the sum of the true positive and false negative predictions, while the FPR is calculated as the number of false positive predictions divided by the sum of the false positive and true negative predictions. The area under the ROC curve (AUC) is a measure of the classifier's overall performance, with a value ranging from 0 to 1, where a higher value indicates better performance.

Another evaluation metric that can be used in classification tasks is the Matthews correlation coefficient (MCC). The MCC is a measure of the correlation between the predicted and actual class labels, with a value ranging from -1 to 1, where a value of 1 indicates perfect prediction, a value of 0 indicates no better than random prediction, and a value of -1 indicates complete disagreement between the predicted and actual labels. The MCC is often used in imbalanced classification tasks, as it is more robust to class imbalance than other metrics, such as accuracy.

It is important to choose the appropriate evaluation metric for a given classification task based on the specific requirements and goals of the task. In some cases, it may be necessary to use multiple evaluation metrics to get a complete picture of the classifier's performance. In this assignment, classification metrics and The effect of the approaches used on the performance of the classifiers in the case of selecting training and test sets are discussed in detail.

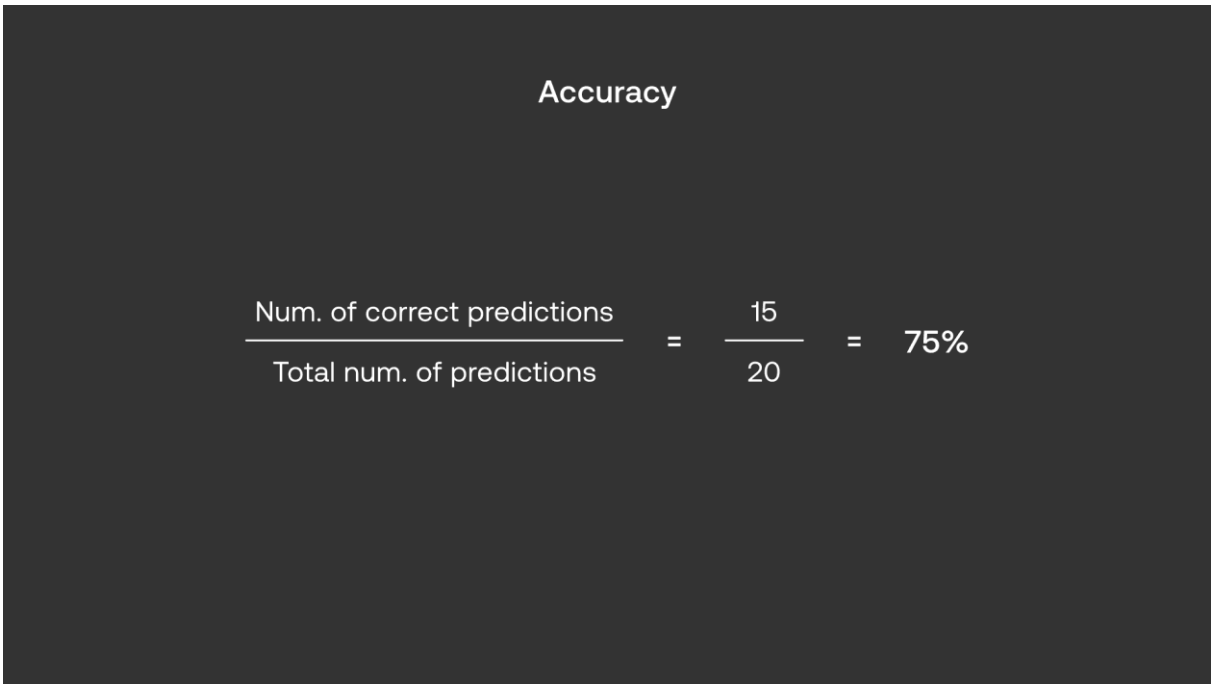
2. CLASSIFICATION EVALUATION METRICS

Binary classification is a type of classification task where there are only two possible classes. An example of binary classification is classifying emails as either spam or not spam. In this example, we have a dataset of 20 email titles that we want to classify using a binary classifier. The classifier will predict whether each email is spam or not spam, and we can then compare the predicted class with the actual class to see how accurate the classifier is. The classifier returns the following outcome:

	Email	Actual	Predicted
1	I need help please wire me $1000 right now	1 - Spam	1 - Spam
2	Hot new investment, don't miss this!	1 - Spam	0 - Not Spam
3	Please help me?	1 - Spam	0 - Not Spam
4	Your parcel will be delivered today	0 - Not Spam	0 - Not Spam
5	Review changes to our Terms and Conditions	0 - Not Spam	0 - Not Spam
6	Weekly sync notes	0 - Not Spam	0 - Not Spam
7	Re: Follow up from today's meeting	0 - Not Spam	0 - Not Spam
8	Pre-read for tomorrow	0 - Not Spam	0 - Not Spam
9	Brief results from new UX study	0 - Not Spam	0 - Not Spam
10	Meeting notes from today	0 - Not Spam	0 - Not Spam
11	A reminder about your next appointment	0 - Not Spam	1 - Spam
12	An invitation to a conference call tomorrow	0 - Not Spam	0 - Not Spam
13	We have some news about your application	0 - Not Spam	1 - Spam
14	Final briefing notes for your meeting	0 - Not Spam	0 - Not Spam
15	Attached are the updated guidelines for this project	0 - Not Spam	0 - Not Spam
16	New feedback on your project from last week's meeting	0 - Not Spam	0 - Not Spam
17	Thank you for participating in our survey	0 - Not Spam	0 - Not Spam
18	Your account has been upgraded to Premium status	0 - Not Spam	1 - Spam
19	Confirming the dates for the annual board meeting	0 - Not Spam	0 - Not Spam
20	Please review these updated guidelines	0 - Not Spam	0 - Not Spam

Şekil 2. 1

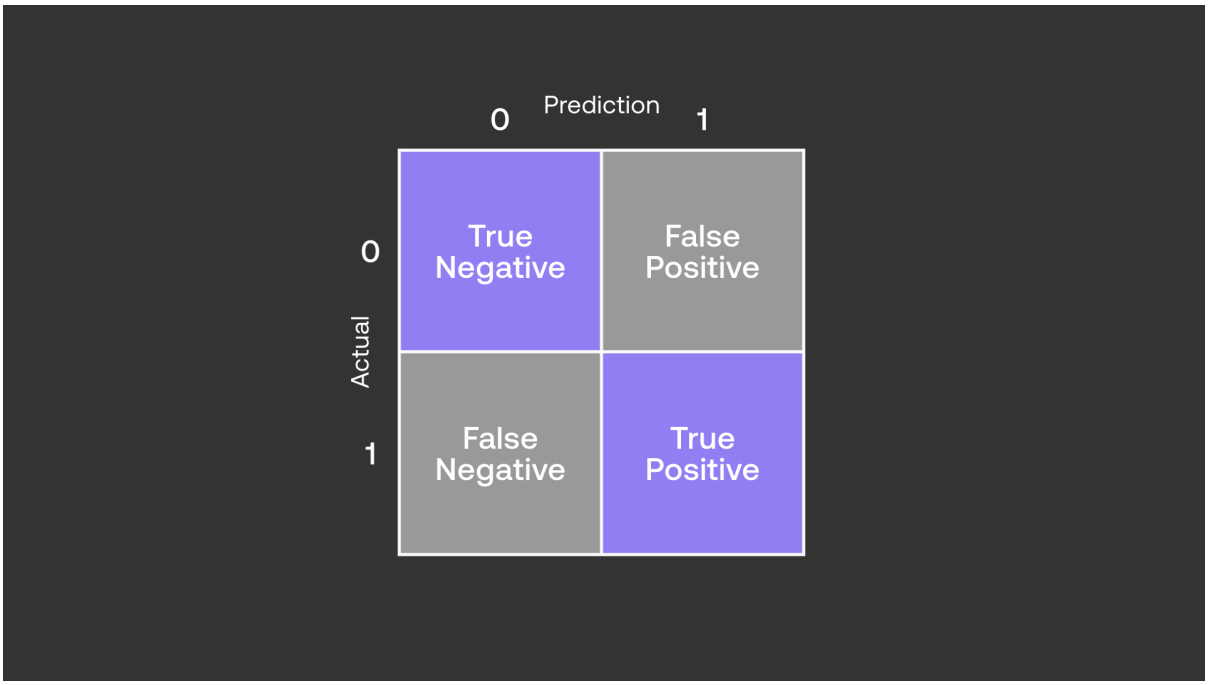
Accuracy is a measure of how well a classifier is performing. It is calculated by comparing the predicted class of each data point with the actual class and counting the number of correct predictions. The accuracy is then calculated by dividing the number of correct predictions by the total number of predictions. For example, if a classifier made 15 correct predictions and 5 incorrect predictions out of a total of 20 predictions, its accuracy would be 75%.



Şekil 2. 2

Accuracy is a commonly used measure for evaluating the performance of a classifier, but it can be misleading in some cases, particularly when the data is imbalanced. Imbalanced data refers to a scenario where some classes have significantly more data points than others. In such situations, a classifier might achieve high accuracy by simply predicting the majority class all the time, even if it is not learning anything useful. For example, if we have a dataset with many more emails that are not spam than emails that are spam, a classifier that always predicts "not spam" would have a high accuracy, even if it is not accurately identifying spam emails. To overcome this issue, it is often necessary to use additional measures of performance beyond accuracy, such as precision and recall.

The confusion matrix is a tool used to evaluate the performance of a classifier. It displays the results of a classification task by grouping them into four categories: true positive (TP), true negative (TN), false positive (FP), and false negative (FN). TP refers to instances where both the actual and predicted values are 1 (e.g. spam emails correctly classified as spam). TN refers to instances where both the actual and predicted values are 0 (e.g. non-spam emails correctly classified as non-spam). FP refers to instances where the actual value is 0 but the predicted value is 1 (e.g. non-spam emails incorrectly classified as spam). FN refers to instances where the actual value is 1 but the predicted value is 0 (e.g. spam emails incorrectly classified as non-spam). The confusion matrix for a binary classification task is a 2-by-2 matrix, where each column represents one class.



Şekil 2. 3 The Confusion Matrix

Applied to our dataset, we get the following values:

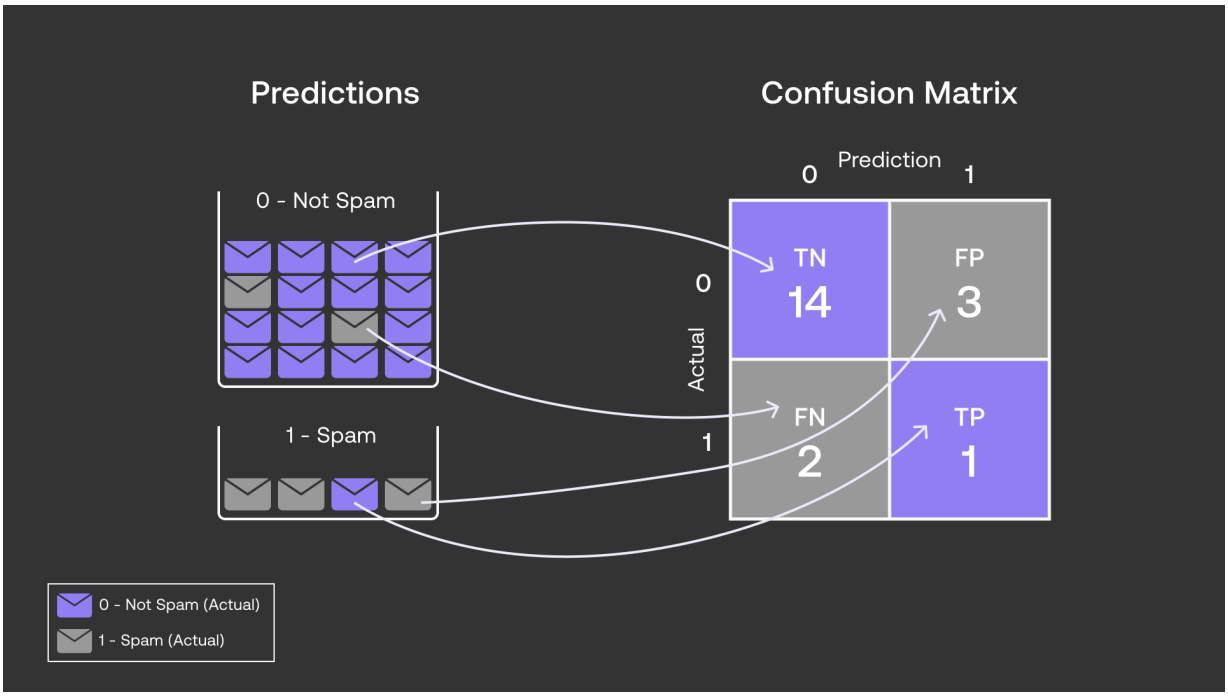
True Positive (TP): 1

True Negative (TN): 14

False Positive (FP): 3

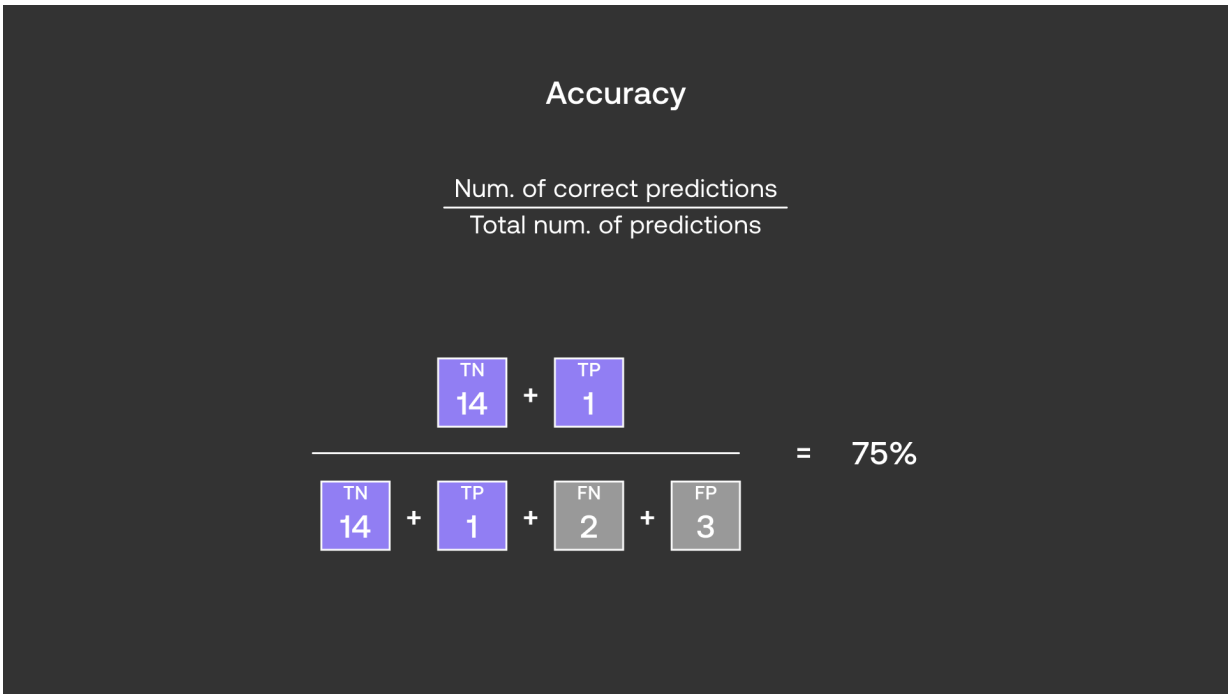
False Negative (FN): 2

We can populate these values in the Confusion Matrix, as follows:



Şekil 2. 4 Populating the classification outcome in the Confusion Matrix

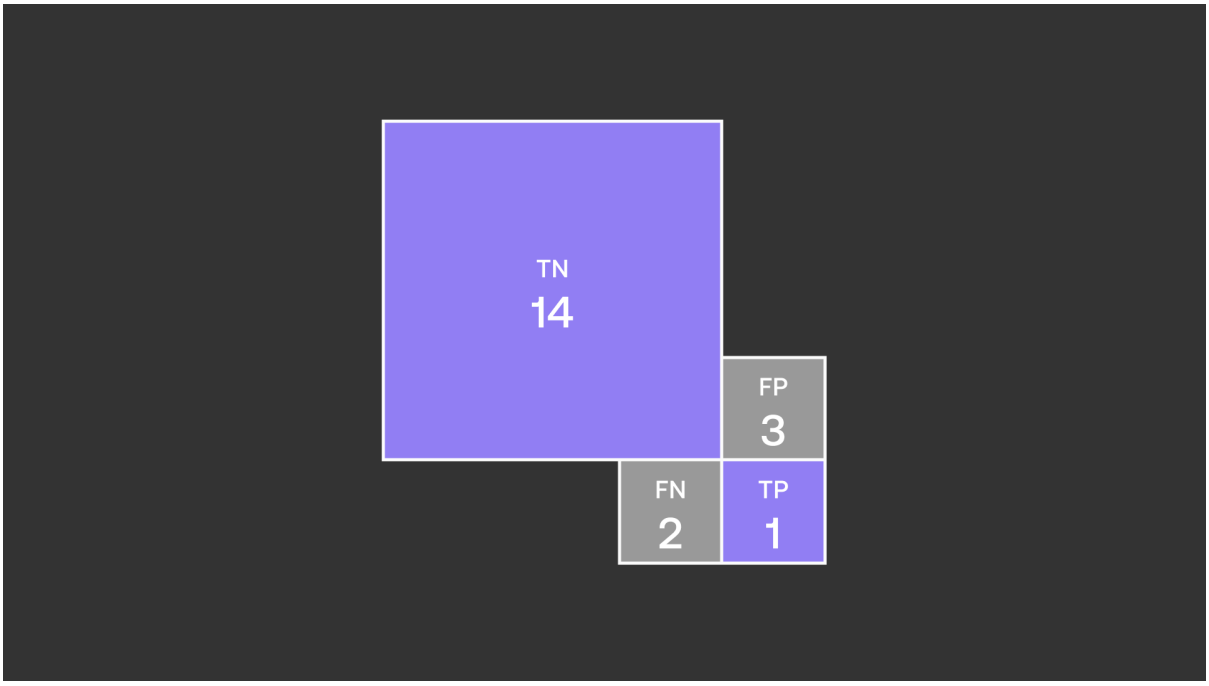
Also map the Confusion Matrix to the Accuracy formula that we saw earlier, as follows:



Şekil 2. 5 Accuracy calculation via the Confusion Matrix

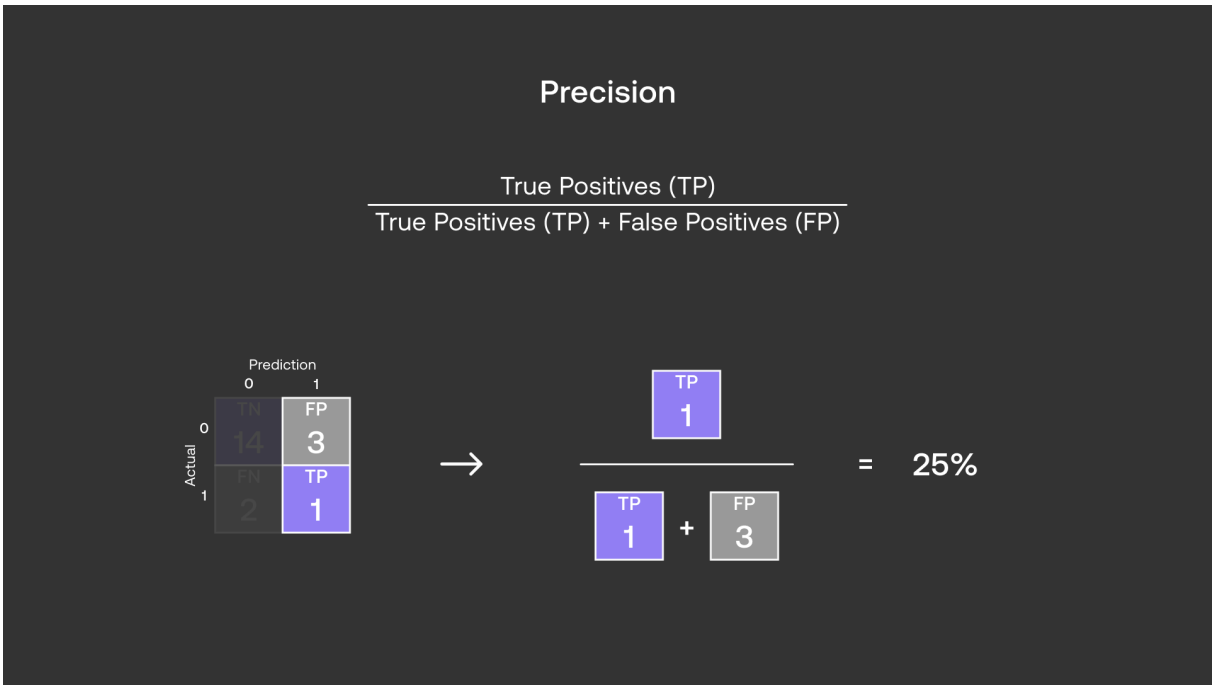
The confusion matrix can be used to understand why accuracy is not always the best measure of classifier performance, especially when dealing with imbalanced datasets. In imbalanced datasets, one class (e.g. non-spam emails) often dominates, meaning that the true negative (TN) category will also dominate the matrix. This can mask the performance of the classifier on the

other classes, making it seem like it is performing better than it actually is. Even if the classifier performs poorly on true positive (TP) and false positive (FP) predictions, for example, its accuracy may still look good because of the high number of TN predictions. This is why it is important to consider other measures of performance, such as precision and recall, in addition to accuracy when evaluating the performance of a classifier on imbalanced datasets.



Şekil 2. 6 The True Negatives dominate the Accuracy calculation

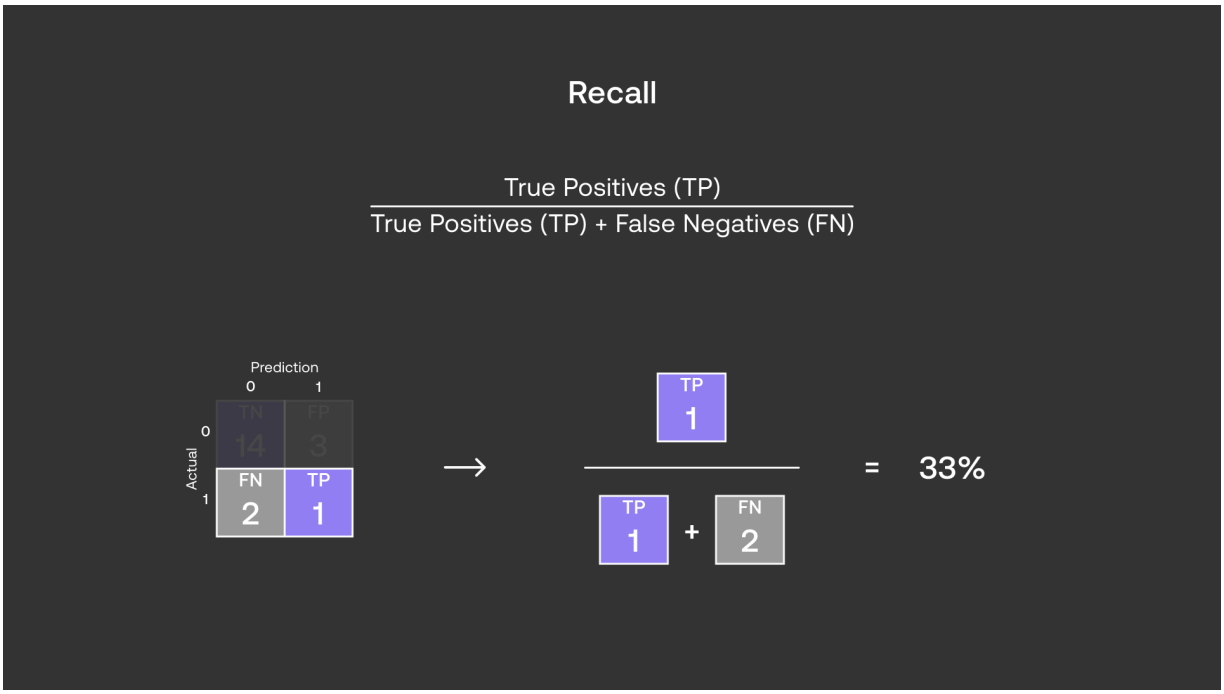
Precision is calculated as follows:



Şekil 2. 7 Precision calculation

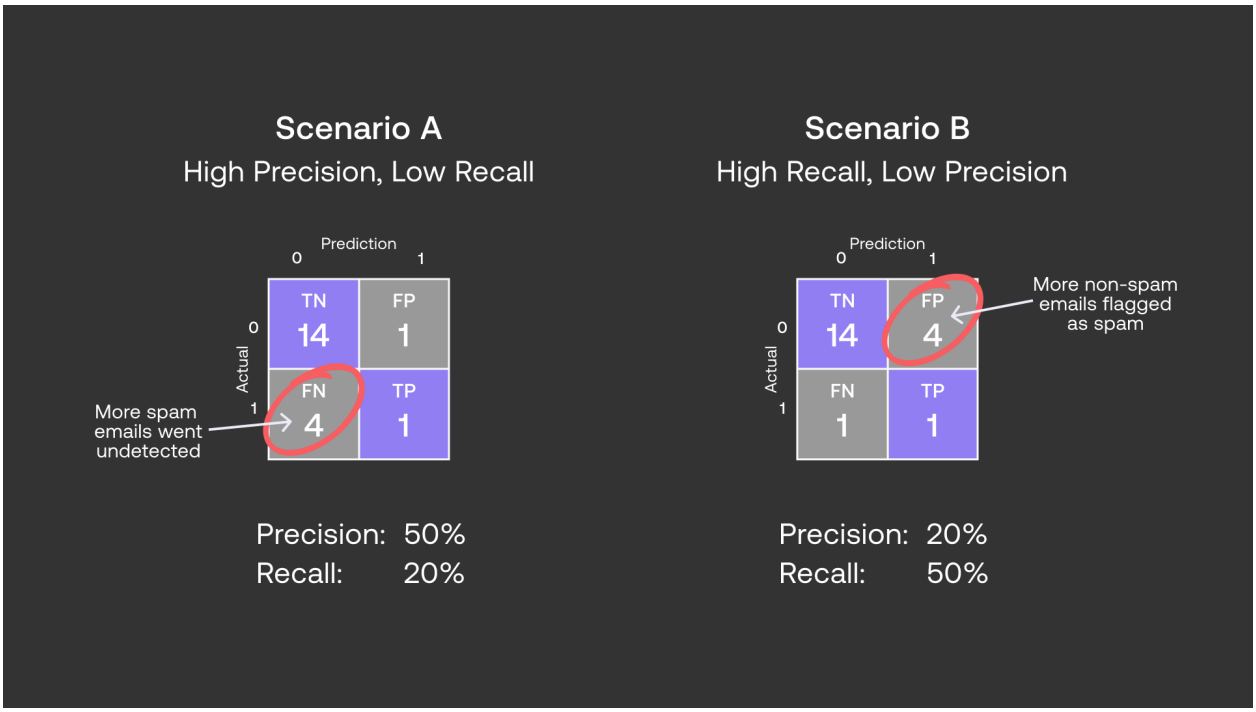
Precision is a measure of a classifier's ability to correctly identify positive instances (e.g. spam emails) out of all the instances it predicted to be positive. It is calculated as the number of true positive predictions divided by the sum of the true positive and false positive predictions. This means that precision only considers the true positive and false positive predictions, and not the true negative or false negative predictions. This can be useful in imbalanced datasets where the true negative class dominates, as it allows us to focus on the performance of the classifier on the minority class (e.g. spam emails). For example, if a classifier has a precision of 25%, it means that out of all the emails it classified as spam, only 25% of them were actually spam emails. This may indicate that the classifier is not very effective at identifying spam emails, even if its overall accuracy looks good. Recall uses the same principle as Precision, except the focus is now on the False Negatives instead of the False Positives. Again, the True Negatives are not part of the consideration.

Recall is calculated as follows:



Şekil 2. 8 Recall calculation

Precision and recall are two measures of a classifier's performance that can be used to evaluate its effectiveness in different ways. Precision is a measure of a classifier's ability to correctly identify positive instances (e.g. spam emails) out of all the instances it predicted to be positive. Recall, on the other hand, is a measure of a classifier's ability to correctly identify all the positive instances (e.g. spam emails) in the dataset. There is a tradeoff between precision and recall, as optimizing for one often comes at the expense of the other. This means that it is difficult to maximize both simultaneously. The appropriate metric to prioritize depends on the nature of the task at hand. For example, in a spam classification task, it may be more desirable to prioritize precision over recall, as this would result in fewer non-spam emails being flagged as spam (i.e. fewer false positives) at the expense of potentially having more spam emails go undetected (i.e. more false negatives). On the other hand, in a task where it is important to identify all positive instances, such as identifying cancerous tumors in medical images, it may be more important to prioritize recall over precision.



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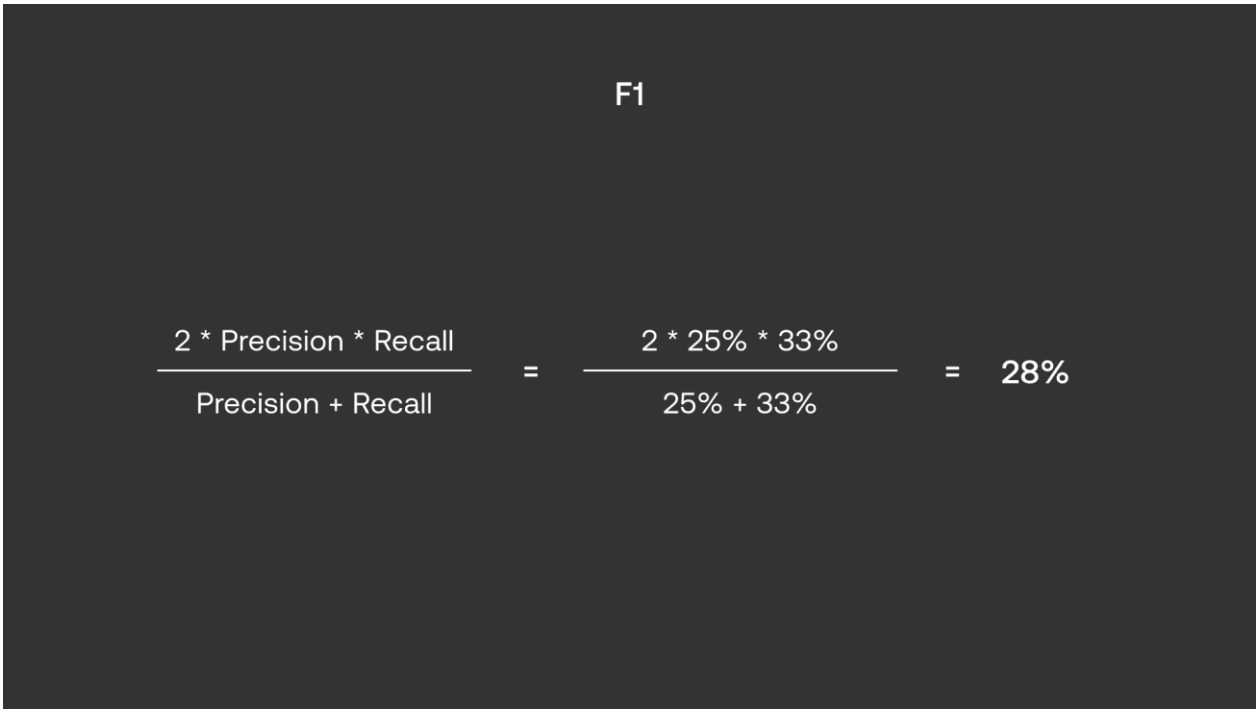
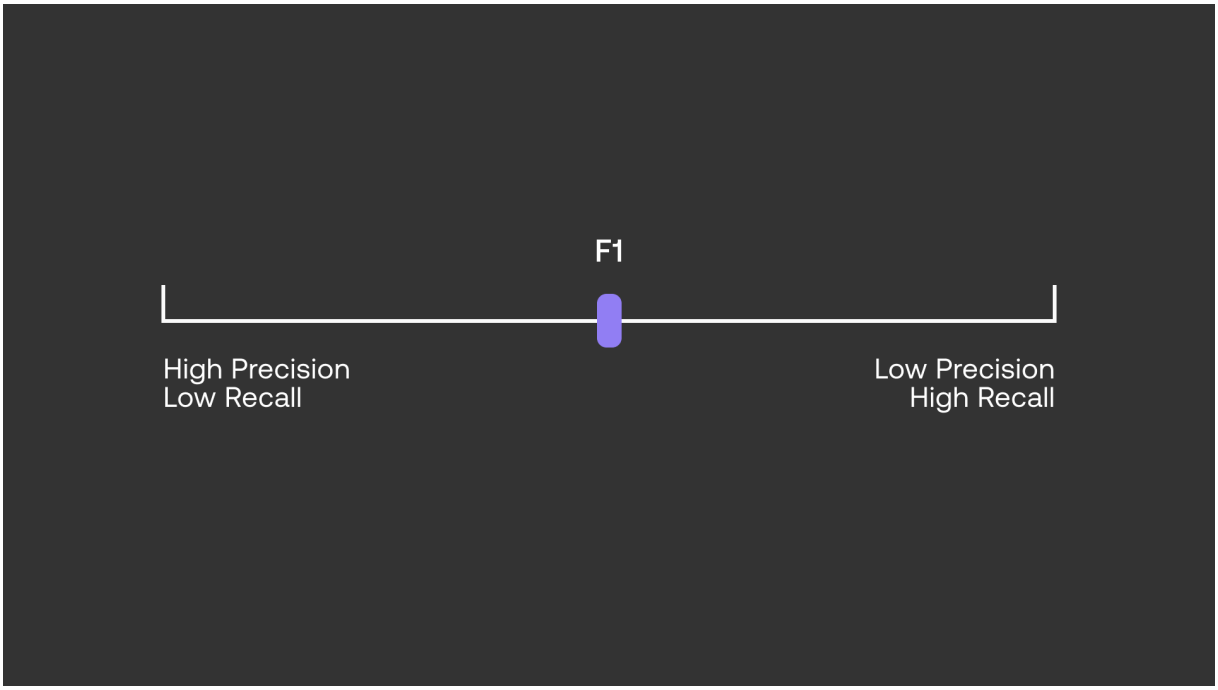
F1 takes into consideration both Precision and Recall. It is calculated as follows:



Şekil 2. 10

F1 score is a measure of a classifier's performance that balances precision and recall. It is calculated as the harmonic mean of precision and recall, with a value ranging from 0 to 1, where a higher value indicates better performance. The F1 score is a good option to use when you

want to balance precision and recall, as it gives equal weight to both metrics. However, there are other versions of the F-score family that you can use if you want to give more weight to one metric over the other. For example, the F2 score gives more weight to recall, while the F0.5 score gives more weight to precision. The appropriate metric to use depends on the specific requirements of the task at hand.



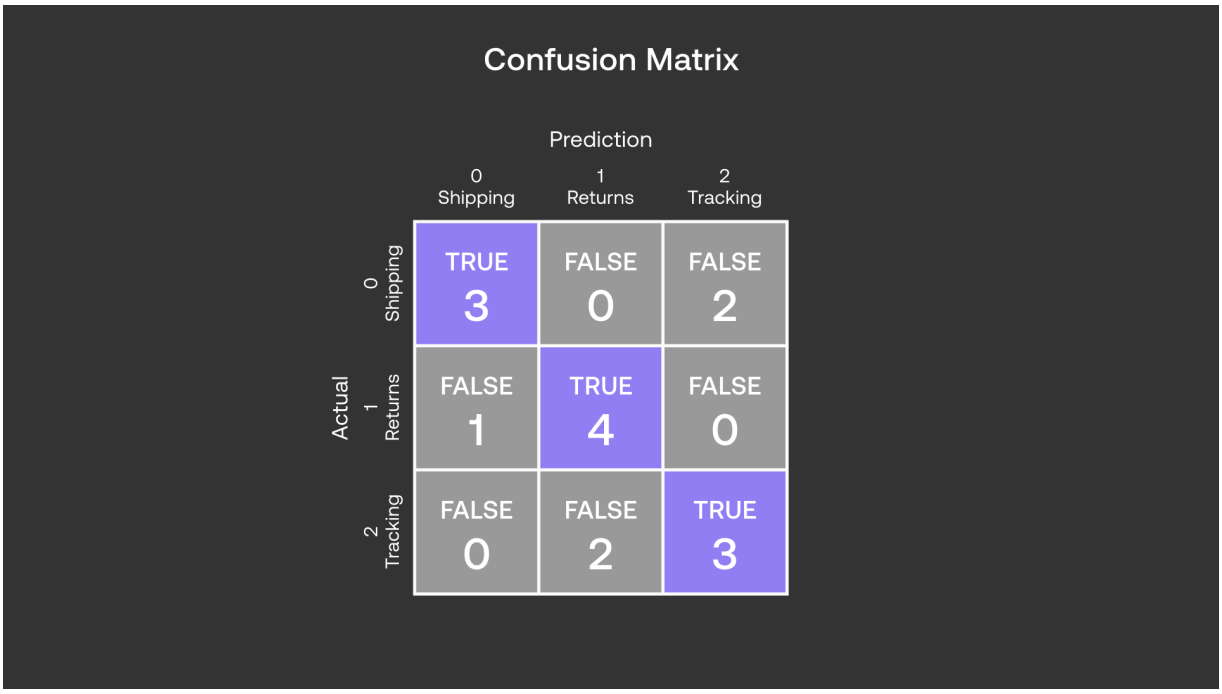
Şekil 2. 11 F1 provides a balance between Precision and Recall

In multi-class classification tasks, where there are more than two classes, the four performance metrics (accuracy, precision, recall, and F1 score) can still be used to evaluate the performance of a classifier. However, they are slightly modified to take into account the additional classes. For example, in a multi-class classification task with three classes (e.g. shipping, returns, and tracking), there will be three different precision values, one for each class, which are calculated as the number of true positive predictions for each class divided by the sum of the true positive and false positive predictions for that class. Similarly, there will be three different recall values, which are calculated as the number of true positive predictions for each class divided by the sum of the true positive and false negative predictions for that class. The F1 score can also be calculated for each class, using the precision and recall values for that class. In this way, the performance of a multi-class classifier can be evaluated and compared across all classes.

	Email	Actual	Predicted
1	Do you offer same day shipping?	0 - Shipping	0 - Shipping
2	Can you ship to Italy?	0 - Shipping	0 - Shipping
3	How long does shipping take?	0 - Shipping	2 - Tracking
4	Can I buy online and pick up in store?	0 - Shipping	0 - Shipping
5	What are your shipping options?	0 - Shipping	2 - Tracking
6	My order arrived damaged, can I get a refund?	1 - Returns	1 - Returns
7	You sent me the wrong item	1 - Returns	1 - Returns
8	I want to exchange my item for another colour	1 - Returns	1 - Returns
9	I ordered something and it wasn't what I expected. Can I return it?	1 - Returns	0 - Shipping
10	What's your return policy?	1 - Returns	1 - Returns
11	Where's my package?	2 - Tracking	1 - Returns
12	When will my order arrive?	2 - Tracking	1 - Returns
13	What's my shipping number?	2 - Tracking	2 - Tracking
14	Which carrier is my package with?	2 - Tracking	2 - Tracking
15	Is my package delayed?	2 - Tracking	2 - Tracking

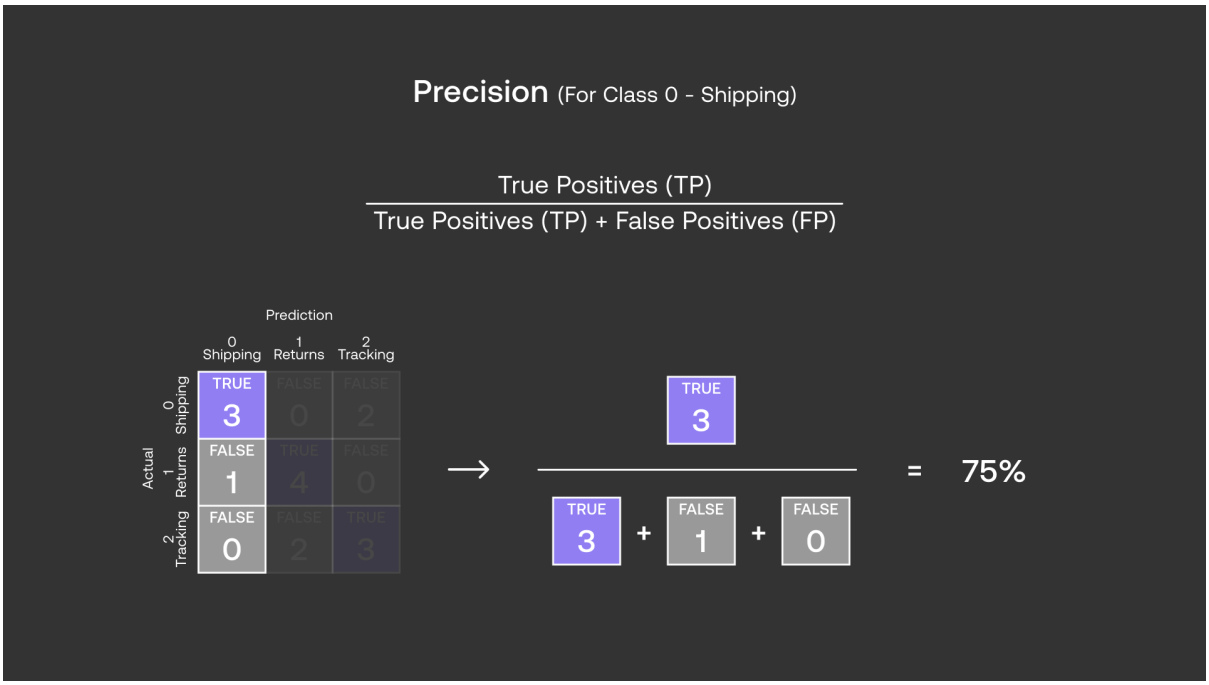
Şekil 2. 12 The email multi-class classification dataset with the actual and predicted classes

In a multi-class classification task, the accuracy of a classifier is calculated in the same way as in a binary classification task: by dividing the number of correct predictions by the total number of predictions. For example, if a classifier made 10 correct predictions and 5 incorrect predictions out of a total of 15 predictions, its accuracy would be 67%. To compute precision, recall, and F1 score for a multi-class classification task, a confusion matrix is constructed. This matrix is a 3-by-3 matrix in the case of a task with three classes, with each column representing one class. The confusion matrix is then used to calculate the precision, recall, and F1 score for each class. For example, in a multi-class classification task with three classes, there will be three different precision values, one for each class, which are calculated using the true positive and false positive predictions for that class. Similarly, there will be three different recall values, which are calculated using the true positive and false negative predictions for each class. The F1 score can also be calculated for each class using the precision and recall values for that class.



Şekil 2. 13 Populating the classification outcome in the Confusion Matrix

In a multi-class classification task, precision and recall are calculated in the same way as in a binary classification task, with the exception that each class will have its own precision and recall values. To calculate precision and recall for a particular class, we can consider that class as the positive class and all the other classes as the negative class. For example, to calculate precision and recall for class 0 (shipping) in a multi-class classification task with three classes, we can consider class 0 as the positive class and classes 1 and 2 as the negative class. Precision for class 0 is then calculated as the number of true positive predictions for class 0 divided by the sum of the true positive and false positive predictions for class 0. Recall for class 0 is calculated as the number of true positive predictions for class 0 divided by the sum of the true positive and false negative predictions for class 0. These values can then be used to calculate the F1 score for class 0 using the formula for F1 score.



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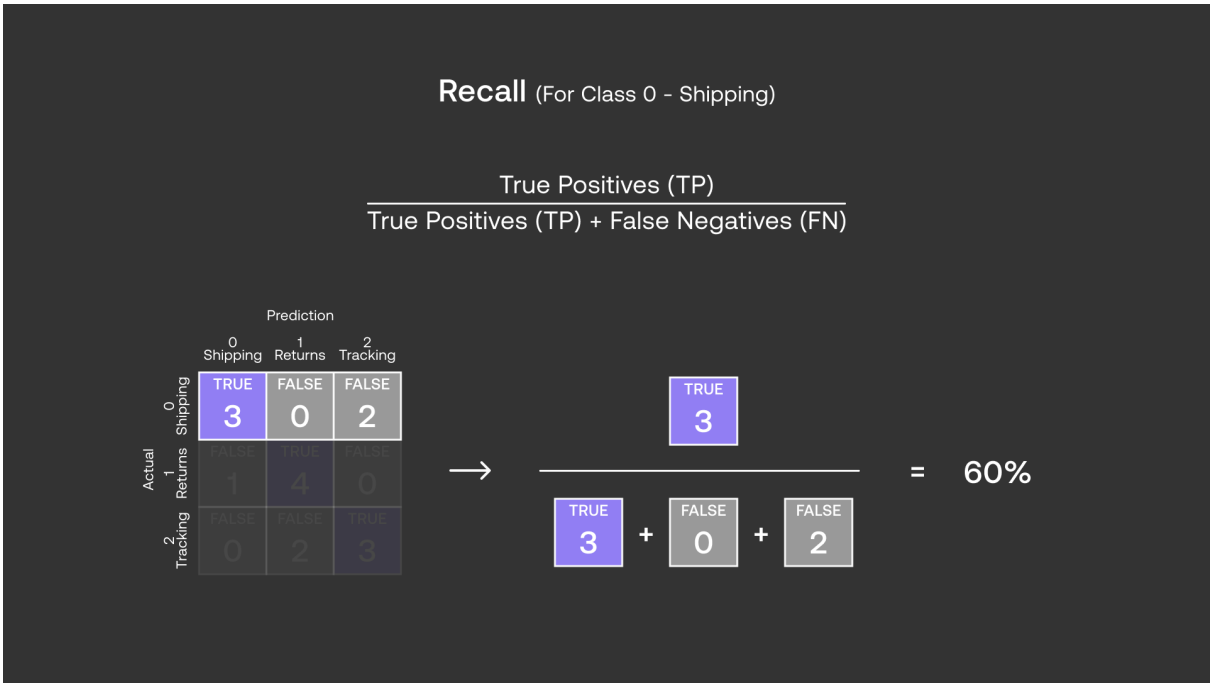
And Recall for class 0 (Shipping) is calculated as follows:



Şekil 2. 15

Each class will have its own F1 too. F1 for class 0 (Shipping) is calculated as follows:

F1 (For Class 0 - Shipping)

$$\frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 * 75\% * 60\%}{75\% + 60\%} = 67\%$$

Şekil 2. 16

Going through all classes in our dataset, this is what Recall, Precision, and F1 look like:

Class	Precision	Recall	F1
0 - Shipping	75%	60%	67%
1 - Returns	67%	80%	79%
2 - Tracking	60%	60%	53%

Şekil 2. 17 **Precision, Recall, and F1 for all three classes**

One way to combine the precision, recall, and F1 score values for each class in a multi-class classification task is to take the average of all the classes. This approach, which is also known as macro-average, involves calculating the average precision, average recall, and average F1

score for all the classes. To do this, we simply sum the precision, recall, or F1 score values for all the classes and divide by the number of classes. This gives us a single value that represents the overall performance of the classifier across all classes. Macro-average can be useful when you want to understand the performance of the classifier on each class separately and see how they compare to one another. However, it can be affected by imbalanced class distributions, as the performance of the classifier on the minority classes may be drowned out by the performance on the majority classes. In such cases, another approach, such as micro-average, may be more appropriate.

Specificity is a performance metric that is used in binary classification tasks to measure the accuracy of the negative class predictions. It is calculated as the ratio of true negative predictions to all negative outcomes, where true negatives are instances that are correctly classified as negative and negative outcomes are instances that are actually negative. This metric is particularly useful in situations where there is a high cost associated with false positive errors (i.e. instances that are incorrectly classified as positive) and it is important to minimize these errors. For example, in a medical diagnosis task, a false positive error (i.e. a healthy patient being diagnosed with a disease) may result in unnecessary treatment and expense, while a false negative error (i.e. a sick patient being misdiagnosed as healthy) may have serious consequences. In such cases, it may be more important to optimize for specificity in order to minimize false positive errors. Specificity is calculated as the number of true negative predictions divided by the sum of the true negative and false positive predictions. A high specificity value indicates that the classifier has a low false positive rate [1,2].

2.1 How Do We Choose?

When choosing an evaluation metric for a classification task, there are several factors to consider. One important factor is the nature of the task itself. For example, in a binary classification task where the goal is to identify positive instances (e.g. spam emails), precision is often a more important metric to consider than accuracy. This is because it is more important to avoid false positive errors (i.e. non-spam emails being classified as spam) than to achieve a high overall accuracy. On the other hand, in a task where the goal is to identify all positive instances, such as identifying cancerous tumors in medical images, recall is often more important than precision, as it is more important to avoid false negative errors (i.e. cancerous tumors being missed) than to achieve a high precision.

Another factor to consider is the class distribution in the dataset. If the dataset is imbalanced, with one class containing significantly more instances than the other class, accuracy may not be the best evaluation metric, as it can be misleading in such cases. In such cases, it may be more appropriate to use a metric such as the F1 score, which balances precision and recall, or the Matthews correlation coefficient (MCC), which is more robust to class imbalance.

Finally, it is important to consider the specific goals and constraints of the task when choosing an evaluation metric. For example, if the goal is to minimize the number of false positive errors, precision may be the most important metric to optimize for. On the other hand, if the goal is to maximize the number of true positive errors, recall may be the most important metric to optimize for. It is also important to consider any constraints or costs associated with false positive or false negative errors, as these can impact the choice of evaluation metric [3].

2.2 Comparing All Metrics

Accuracy, precision, recall, and F1 score are all evaluation metrics that can be used to measure the performance of a classifier. However, each of these metrics has its own characteristics and is most appropriate for certain types of classification tasks.

Accuracy is the most straightforward evaluation metric and is calculated as the number of correct predictions divided by the total number of predictions. It is a good overall measure of classifier performance, but can be misleading in imbalanced datasets, where one class contains significantly more instances than the other class.

Precision is a measure of the classifier's ability to avoid false positive errors. It is calculated as the number of true positive predictions divided by the sum of the true positive and false positive predictions. Precision is particularly important in tasks where false positive errors are costly or undesirable, such as in spam classification or medical diagnosis.

Recall is a measure of the classifier's ability to identify all positive instances. It is calculated as the number of true positive predictions divided by the sum of the true positive and false negative predictions. Recall is particularly important in tasks where false negative errors are costly or undesirable, such as in fraud detection or disease diagnosis. F1 score is a balance between precision and recall, and is calculated as the harmonic mean of precision and recall. It is a good overall measure of classifier performance and is often used when it is important to balance precision and recall.

3. THE EFFECT OF THE APPROACHES USED ON THE PERFORMANCE OF THE CLASSİFİERS İN THE CASE OF SELECTİNG TRAİNİNG AND TEST SETS

In this article, the authors examine the effect of different training and testing approaches on the performance of machine learning models in biomedical datasets. Biomedical datasets are datasets that are used in medical research and clinical practice, and often contain sensitive personal information about patients. These datasets can be challenging to work with due to factors such as class imbalance (where one class contains significantly more instances than the other class) and the need to protect patient privacy.

The authors discuss various approaches to training and testing machine learning models, including cross-validation and stratified sampling. Cross-validation is a method of evaluating the performance of a machine learning model by dividing the dataset into training and testing sets and training the model on the training set and evaluating it on the testing set. Stratified sampling is a method of sampling from a dataset in which the proportion of instances in each class is preserved in the sample. The authors compare the effectiveness of these approaches in terms of model performance, using several evaluation metrics such as accuracy, precision, and recall. They also discuss the challenges and considerations involved in working with biomedical datasets, including the need to handle class imbalance and protect patient privacy. Overall, the authors conclude that the choice of training and testing approach can significantly impact the performance of machine learning models in biomedical applications. They recommend the use of appropriate evaluation metrics and techniques to ensure the reliability and validity of the results. In order to more fully understand the extent to which different training and testing approaches can affect the performance of classifiers, it is necessary to conduct a controlled study in which the same classifiers are trained and tested using a range of different approaches. This can be done by first selecting a number of datasets with different characteristics, such as imbalanced datasets, high-dimensional datasets, etc., and then using these datasets to train and test a range of classifiers using different approaches. One common approach to training and testing classifiers is cross-validation, in which the dataset is divided into a training set and a testing set, and the classifier is trained on the training set and evaluated on the testing set. This process is repeated a number of times, with different portions of the dataset being used as the testing set each time, in order to get a more robust estimate of the classifier's performance. Another approach is stratified sampling, in which the proportion of instances in each class is preserved in the training and testing sets, in order to ensure that the classifier is exposed to a representative sample of the data [4].

In order to compare the performance of the classifiers using different training and testing approaches, a range of evaluation metrics can be used, such as accuracy, precision, and recall. These metrics provide different perspectives on the classifier's performance, and can help to identify the strengths and weaknesses of the different approaches.

It is also important to consider factors such as the size of the training and testing sets, as well as the specific algorithms and parameters used by the classifiers, in order to ensure that the results of the study are robust and generalizable. By considering these factors, it is possible to more fully understand the extent to which different training and testing approaches can affect the performance of classifiers, and to identify the most effective approach for a given task.

Hold-out set method:

The hold-out set method is a simple and widely used method for evaluating the performance of a classifier. It involves dividing the dataset into a training set and a testing set, with the training set being used to train the classifier and the testing set being used to evaluate its performance. The advantage of this method is that it is relatively simple to implement, and it allows for a clear separation of the training and testing phases. However, the hold-out set method has some limitations. One major limitation is that the performance of the classifier can be sensitive to how the data is divided into the training and testing sets. If the training and testing sets are not representative of the overall dataset, the classifier's performance may be overestimated or underestimated. This can lead to unreliable results and can make it difficult to compare the performance of different classifiers.

Multiple random sampling:

Multiple random sampling is a method that involves dividing the dataset into training and testing sets multiple times and training and evaluating the classifier on each set. The performance of the classifier is then averaged across the different splits. This method can provide a more robust estimate of the classifier's performance, as it accounts for the variability in the training and testing sets. However, multiple random sampling can be time-consuming to implement, especially if the dataset is large. It also requires a large number of splits in order to get a reliable estimate of the classifier's performance, which can be computationally expensive. Cluster sampling is a sampling method that involves dividing the population into groups or clusters, and then selecting a sample from these clusters. One of the main advantages of cluster sampling is that it allows researchers to apply random or cluster sampling after determining the groups. This method is useful for researchers who are collecting primary data from a

geographically dispersed population, as it allows them to divide the population into groups without any restrictions. Additionally, cluster sampling is cost-effective and time-effective, as it helps researchers to cut down the population into smaller groups. This method is also convenient for researchers, as it does not require a complete list of all the members of the target population, which dramatically reduces sample preparation costs. Finally, cluster sampling allows researchers to choose their sample carefully, as they can mindfully select the audience, which can help to decrease any issues that may arise during random sampling.

K-fold cross validation:

k-fold cross validation is a widely used method for evaluating the performance of a classifier. It involves dividing the dataset into k equal-sized folds and training and evaluating the classifier k times, using a different fold as the testing set each time. The performance of the classifier is then averaged across the k folds. This method is relatively efficient, as the classifier is trained and evaluated only k times. It also provides a more robust estimate of the classifier's performance, as it accounts for the variability in the training and testing sets. However, it is still sensitive to the choice of k and the way in which the folds are constructed. If k is too small, the estimate of the classifier's performance may be unreliable, and if k is too large, the computation may be slow. Ability to detect overfitting: Cross-validation allows you to assess whether your model is overfitting the training data by measuring its performance on multiple different test sets. Improved model selection: Cross-validation can be used to compare the performance of different models on the same dataset, helping you choose the best model for your problem.

Ability to handle small datasets: When you have a small dataset, using a hold-out set for model evaluation can result in a high variance in the model's performance. Cross-validation allows you to use all the data for training and evaluation, reducing the variance in the model's performance. Insights into model performance: Cross-validation provides a distribution of model performance, which can give you insights into how well your model is likely to perform on unseen data. Improved model performance: Finally, the author suggests that using cross-validation can lead to improved model performance, as it allows you to tune hyperparameters more effectively. In conclusion, it is important to carefully consider the evaluation method when evaluating the performance of a classifier, as it can have a significant impact on the results.

Bootstrap:

The bootstrap method is a resampling technique that involves drawing a number of random samples with replacement from the original dataset and training and evaluating the classifier on each sample. The performance of the classifier is then averaged across the samples.

This method can provide a more robust estimate of the classifier's performance, as it accounts for the variability in the training and testing sets. It is also relatively efficient, as the classifier is trained and evaluated only a limited number of times. However, it can be sensitive to the choice of the number of samples and the sampling method. It is important to note that the choice of performance metric will depend on the specific requirements and goals of your classification task. For example, if you are more concerned with minimizing false negatives, you may prioritize recall over precision [5,6,7,8].

8.CONCLUSION

It is not possible for me to write a conclusion for your homework without more context about the specific task and data that you are working with. However, I can provide some general guidance on evaluating the performance of a classifier.

When evaluating the performance of a classifier, it is important to choose an appropriate evaluation method that takes into account the characteristics of your data and the requirements of your task. Some common evaluation methods include the hold-out set method, multiple random sampling, k-fold cross validation, and the bootstrap method. Each of these methods has its own advantages and limitations, and it is important to consider these when choosing the most appropriate method for your task. For example, the hold-out set method is simple to implement, but it can be sensitive to the way the data is divided into training and testing sets. Multiple random sampling can provide a more robust estimate of the classifier's performance, but it is time-consuming to implement. K-fold cross validation is efficient and provides a robust estimate of the classifier's performance, but it is sensitive to the choice of k. The bootstrap method is relatively efficient and provides a robust estimate of the classifier's performance, but it can be sensitive to the sampling method. One of the main advantages of bootstrapping is that it is a simple and convenient way to estimate the standard errors and confidence intervals of a statistic. This method avoids the cost and time associated with repeating an experiment to obtain additional samples, as it allows researchers to use the original sample to generate multiple simulated samples. Additionally, bootstrapping is a useful method for estimating the population parameters when the underlying distribution of the data is unknown or when the sample size is

small. Overall, bootstrapping is a powerful tool for assessing the reliability of statistical estimates and for making inferences about the population based on a sample.

REFERENCES

[1] "Classification Evaluation Metrics," Cohere AI, [Online]. Available: https://txt.cohere.ai/classification-eval-metrics/. [Accessed: 28-Dec-2022].

[2] S. V. Kaggle, "Understanding Common Classification Metrics: Titanic Style," Towards Data Science, [Online]. Available: https://towardsdatascience.com/understanding-common-classification-metrics-titanic-style-8b8a562d3e32. [Accessed: 28-Dec-2022].

[3] "Multistage Sampling: Advantages and Application," QuestionPro, [Online]. Available: https://www.questionpro.com/blog/multistage-sampling-advantages-and-application. [Accessed: 28-Dec-2022].

[4] Uçar, Muhammed & Nour, Majid & Sindi, Hatem & Polat, Kemal. (2020). The Effect of Training and Testing Process on Machine Learning in Biomedical Datasets. Mathematical Problems in Engineering. 2020. 1-17. 10.1155/2020/2836236.

[5] "Holdouts and Cross-Validation: Why the Data Used to Evaluate your Model Matters," Alteryx Community, [Online]. Available: https://community.alteryx.com/t5/Data-Science/Holdouts-and-Cross-Validation-Why-the-Data-Used-to-Evaluate-your/ba-p/448982. [Accessed: 28-Dec-2022].

[6] "Hold-out Method for Training Machine Learning Model," VitalFlux, [Online]. Available: https://vitalflux.com/hold-out-method-for-training-machine-learning-model. [Accessed: 28-Dec-2022].

[7] "What Is Bootstrapping in Statistics?" Built In, [Online]. Available: https://builtin.com/data-science/bootstrapping-statistics. [Accessed: 28-Dec-2022].

[8] Nitesh V. Chawla and Grigoris Karakoulas. 2005. Learning from labeled and unlabeled data: an empirical study across techniques and domains. J. Artif. Int. Res. 23, 1 (January 2005), 331–366.